**Search History saved as "2020.11. SR_volatile anesthetics vs intravenous sedation_cognition_M_FNL"**
**Search History saved as "2020.11. SR_volatile anesthetics vs intravenous sedation_cognition_E_FNL"**
**Search History saved as "2020.11. SR_volatile anesthetics vs intravenous sedation_cognition_PsycInfo_FNL"**

***Topic***: Effect of Sedation with Inhaled Anesthetics on Cognitive and Psychiatric Outcomes in Critically Ill Adults: A Systematic Review

***For:*** Dr. Sean Cuninghame; Dr. Kevin Gorsky, Dr. Conall Franceour, Dr. Davinia Withington, Dr. Lisa Burry, Dr. Angela Jerath, Dr. Marat Slessarev,

**Database: Ovid MEDLINE(R) ALL <1946 to December 02, 2020>**

Search Strategy:

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1 exp Anesthetics, Inhalation/ or Anesthesia, Inhalation/ (60387)

2 (((inhal* or insufflation$ or gas or gaseous$ or volatile$ or halogen$) adj3 (an?esthe* or sedat*)) or (inhalation$ adj3 induction$) or (ftorotan$ or fluothane$ or narcotan$ or isoflurane$ or aerane$ or aerrane$ or forane$ or forene$ or forthane$ or isoflurano$ or isorane$ or sofloran$ or sevocalm$ or sevoflo$ or sevofrane$ or sevohale$ or sevoflurane$ or sevorane$ or sevotec$ or ultane$ or anestane$ or bromochlorotrifluorethane$ or (bromo$ adj2 chloro$ adj2 trifluorethane$) or fluorothan* or halan$ or halothan* or halthane$ or ineltano$ or phthorothane$ or trothane$ or desflurane$ or enflurane$ or penthrox$)).tw,kf. (42998)

3 or/1-2 [volatile anesthetics concept] (76374)

4 Critical Care/ or Critical Illness/ or exp Intensive Care Units/ (145037)

5 ((intens* or critical*) adj3 (care or ill or illness or unit*)).tw,kf. (212397)

6 (ICU or ICUs or CCU or CCUs or ITU or HDU).tw,kf. (65415)

7 or/4-6 [ICU concept] (276089)

8 (intens$ adj3 care).af. or (intensive* or critical$).jw,ja,jn. or (sedat$.tw,kf. and (exp *Respiration, Artificial/ or (((ventilat$ or respirat$) adj2 (artificial$ or mechanical$ or pneumon$ or controlled)) or (ventilat$ adj3 patient$)).tw,kf.)) (360980)

9 7 or 8 (429901)

10 3 and 9 (4309)

11 Anesthesia, Intravenous/ or Infusions, Intravenous/ or (intravenous$ or intra-venous$).mp. or (IV or IVs or TIVA or Auto?TIVA or Manual?TIVA).tw,kf. (804110)

12 Propofol/ or (propofol$ or diprivan$).tw,kf. or Ketamine/ or (ketamin$ or esketamin$ or ketanest$ or ketalar$ or ketaset$).tw,kf. or Midazolam/ or (midazolam$ or dormicum$ or Versed$ or buccolam$ or dalam or dormonid or fortanest or fulsed or hypnovalor hypnovel or hypnoyvel or ipnovel or midacum or midazo or midazol or midolamor miloz).tw,kf. or conscious sedation/ or deep sedation/ or ((conscious$ or moderate$ or deep$) adj2 sedat$).tw,kf. (62360)

13 12 and (continuous$ or prolonged$ or concentration$ or infusion$).tw,kf. (18739)

14 11 or 13 [Intravenous Anesthesia OR Midazolam/Ketamine/Propofol/ continuous administration - concept] (813003)

15 10 and 14 (1475)

16 exp Cognition Disorders/ or exp Cognition/ or exp Cognitive Therapy/ or Cognitive Aging/ or Tremor/ or exp neuropsychological tests/ (409458)

17 neurobehavioral manifestations/ or exp communication disorders/ or exp memory disorders/ or intellectual disability/ or exp perceptual disorders/ or exp sensation disorders/ (355136)

18 (cognition$ or cognitive$ or neuropsycholog$ or neuro-psycholog$ or neurocognit$ or neuro?affectiv$ or neuro-cognit$ or neurobehav$ or neuro-behav$ or aphasia$ or (involuntary$ adj2 quiver$) or tremor$).mp. (575907)

19 exp *Multiple Sclerosis/ or *Alzheimer Disease/ or exp *parkinsonian disorders/ or delirium/ or (delirium$ or delirious$ or Parkinson$ or Alzheimer$).tw. or ((multiple$ or disseminated$) adj5 sclerosis$).tw,kw. (358897)

20 or/16-19 (1269102)

21 exp Prefrontal Cortex/pa, pp or exp Cerebral Cortex/pa, pp or exp Hippocampus/pa, pp or Cerebral Infarction/pa, pp or White Matter/pa, pp or Brain Diseases/pa, pp or Subthalamic Nucleus/pa, pp or exp Subarachnoid Hemorrhage/pa, pp (128097)

22 (hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-thalam$ or intracranial$ or intra-cranial$).tw,kw. (1015992)

23 ((cortex$ or cortices$ or area$ or region$) adj3 (orbital$ or prefrontal$ or orbitofrontal$)).tw,kw. (54920)

24 ((pars$ adj (triangulari$ or operculari$)) or ((area$ or region$) adj2 (speech$ or language$ or broca$)) or broca's or brocas or subcallosal$).tw,kw. (6116)

25 ((frontal$ or olfactory$) adj3 (sulci or (superior$ adj5 convolution$) or sulcus$)).tw,kw. (665)

26 ((frontal$ or orbital$ or orbitofrontal$ or rectal or rectus or straight or marginal) adj3 (gyri$ or gyrus$)).tw,kw. (11762)

27 or/21-26 (1115455)

28 (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$).tw,kw. (1691979)

29 27 and 28 (291070)

30 ((brain or cortex$ or cortices$ or cortical$ or subcortical$ or sub-cortical$ or hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-

thalam$ or intracranial$ or intra-cranial$) adj2 (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$ or impair$ or decline$ or defect$ or lesion$ or disorder$ or patholog$ or integrity or hyperintens$)).tw,kw. (256693)

31 ((visuospatial$ or (visuo$ adj2 spatial$) or (verbal$ adj2 (comm$ or fluen$)) or (process$ adj2 speed$) or psychomotor$ or psycho-motor$ or memory$ or personality$ or perceptual$ or attention$ or concentration$ or auditory$ or visual$ or sensory$ or sensation$ or spatial$ or IQ or intelligence$ or intellectual$) adj5 (impairment$ or impaired or deficit$ or deficien$ or abnormal$ or dysfunction$ or function$ or skill$ or problem$ of complain$)).mp. (285691)

32 (MMSE or (mini?mental$ adj3 stat$ adj3 exam$)).tw,kf. (11443)

33 exp Stress, Psychological/ or anxiety/ or Depression/ or depressive disorder/ or depressive disorder, major/ or depressive disorder, treatment-resistant/ or (stress$ or ((hospital or scale$ or score$ or questionnaire$) and (HADS-D or PHQ-9 or PHQ or HADS)) or depress$ or anxiety$).tw,kf. (1471121)

34 Stress Disorders, Post-Traumatic/ or (shock$ adj2 shell$).tw,kf. or ((posttrauma$ or (post$ adj3 traumatic$) or post-trauma$) adj3 (stress$ or disorder$ or psychiatric$)).tw,kf. or ((acute$ adj2 stress$ adj2 disorder$) and ASD).tw,kf. (47060)

35 ((posttraumatic$ or (post$ adj3 traumatic$) or post-traumatic$ or stress$ or disorder$) and PTSD).tw,kf. (24021)

36 or/33-35 (1478446)

37 blood pressure/ or heart rate/ or ((heart$ or pulse$) adj3 rate$1).tw. or (cardiac$ adj2 chronotrop$).tw. or (pressure$ adj3 (blood$ or diastolic$ or pulse$ or systolic$)).tw. (611661)

38 (psychiatr$ or disability$).tw,kf. or exp Intellectual Disability/ or exp Disability Evaluation/ or "International Classification of Functioning, Disability and Health"/ (522184)

39 or/37-38 (1128539)

40 20 or 29 or 30 or 31 or 32 or 36 or 39 [Cognitive concept ] (3749930)

41 10 and 40 [ICU concept PLUS volatile anesthetics concept PLUS Cognitive concept ] (1299)

42 exp Survivors/ or surviv$.mp. or (post?intensive$ or postdischarg$ or ((post$ or after$) adj3 (critical$ or ICU or intensive$ or discharg$))).tw,kf. or Comparative Study/ or (vs$1 or vs or versus or compar$ or during-and-after).tw. or ((multicenter$ or multi-center$).tw. and ((between or each or both or either or than or across) adj2 group$1).tw,kw.) or groups.tw. or group$1.ab. /freq=2 (9075813)

43 ((random$ or multicenter$ or multi-center$ or center$ or multicentr$ or multi-centr$ or centr$ or pilot$ or population$ or observational or validation or phase$) adj3 (trial or study)).mp. or clinical trial.tw. or exp clinical trial/ or double-blind method/ or random allocation/ (1809348)

44 or/42-43 [ Random/Clinical trial / Survivors / Comparative Study] (9679756)

45 10 and 14 and 44 [WITHOUT Cognitive concept PRESENT could be found in the Full Text - ICU concept PLUS volatile anesthetics concept PLUS Intravenous Anesthesia concept PLUS Random/Clinical trial / Survivors / Comparative Study] (1095)

46 41 or 45 (1915)

47 46 not (exp Animals/ not (Human/ and exp Animals/)) (1511)

48 47 not (animal$1 or mice or rat or rats or cat$1 or cattle$1 or dog$1 or goat$1 or horse$1 or rabbit$1 or sheep$1 or swine$1 or pig$1 or piglet* or canine$1 or feline$1 or porcine$ or calf or primate* or rodent$ or hamster$ or lamb$1 or monkey$1 or murine or veterinar*).ti. (1480)

49 limit 48 to "all adult (19 plus years)" (863)

50 limit 48 to "all child (0 to 18 years)" (425)

51 48 not (50 not (49 and 50)) (1205)

52 51 and adult$.ti. (32)

53 51 and (pediatr$ or paediatr$ or child$ or adolescent$ or infan$ or newborn$ or boy$1 or neonat$).ti. (23)

54 51 not (53 not (52 and 53)) (1184)

55 exp case-control studies/ or (case$ and control$).tw. or (case$ and series).tw. [Medline case series ] (1623777)

56 case reports/ or case report$.mp. (2227506)

57 54 not (56 not (55 and 56)) (1117)

58 57 not case report.ti. [Removing case Reports and retaining Case Series] (1113)

59 limit 58 to english language (1038)

**60 limit 59 to yr="1970 -Current" (1037)**

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**Database: Embase Classic+Embase <1947 to 2020 December 02>**

Search Strategy:

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1 exp *inhalation anesthetic agent/ or exp inhalation anesthesia/ (73272)

2 (((inhal* or insufflation$ or gas or gaseous$ or volatile$ or halogen$) adj3 (an?esthe* or sedat*)) or (inhalation$ adj3 induction$) or (ftorotan$ or fluothane$ or narcotan$ or isoflurane$ or aerane$ or aerrane$ or forane$ or forene$ or forthane$ or isoflurano$ or isorane$ or sofloran$ or sevocalm$ or sevoflo$ or sevofrane$ or sevohale$ or sevoflurane$ or sevorane$ or sevotec$ or ultane$ or anestane$ or bromochlorotrifluorethane$ or (bromo$ adj2 chloro$ adj2 trifluorethane$) or fluorothan* or halan$ or halothan* or halthane$ or ineltano$ or phthorothane$ or trothane$ or desflurane$ or enflurane$ or penthrox$)).tw,kw. (59537)

3 or/1-2 [volatile anesthetics concept] (100064)

4 critically ill patient/ or exp intensive care unit/ or intensive care/ or critical illness/ (351059)

5 ((intens* or critical*) adj3 (care or ill or illness or unit*)).tw,kw. (322355)

6 (ICU or ICUs or CCU or CCUs or ITU or HDU).tw,kw. (133390)

7 or/4-6 [ICU concept] (483378)

8 (intensive* or critical$).ja,jn,jx. or (sedat$.tw,kw. and (exp *artificial ventilation/ or (((ventilat$ or respirat$) adj2 (artificial$ or mechanical$ or pneumon$ or controlled)) or (ventilat$ adj3 patient$)).tw,kw.)) (267016)

9 7 or 8 (640893)

10 3 and 9 (4189)

11 exp intravenous anesthesia/ or exp intravenous drug administration/ or (intravenous$ or intra-venous$).mp. or (IV or IVs or TIVA or Auto?TIVA or Manual?TIVA).tw,kw. (1724896)

12 Propofol/ or (propofol$ or diprivan$).tw,kw. or Ketamine/ or (ketamin$ or esketamin$ or ketanest$ or ketalar$ or ketaset$).tw,kw. or Midazolam/ or (midazolam$ or dormicum$ or Versed$ or buccolam$ or dalam or dormonid or fortanest or fulsed or hypnovalor hypnovel or hypnoyvel or ipnovel or midacum or midazo or midazol or midolamor miloz).tw,kw. or conscious sedation/ or deep sedation/ or ((conscious$ or moderate$ or deep$) adj2 sedat$).tw,kw. (137818)

13 12 and (continuous$ or prolonged$ or concentration$ or infusion$).tw,kw. (34730)

14 11 or 13 [Intravenous Anesthesia OR Midazolam/Ketamine/Propofol/ continuous administration - concept] (1741681)

15 10 and 14 (1848)

16 exp *cognitive defect/ or exp *Cognition/ or exp *Cognitive Therapy/ or exp *neuropsychological test/ or experimental cognitive test/ (1135100)

17 exp *communication disorder/ or exp *memory disorders/ or exp *perception disorder/ or exp *sensory dysfunction/ (288272)

18 (cognition$ or cognitive$ or neuropsycholog$ or neuro-psycholog$ or neurocognit$ or neuro?affectiv$ or neuro-cognit$ or neurobehav$ or neuro-behav$ or aphasia$ or (involuntary$ adj2 quiver$) or tremor$).mp. (866016)

19 *Multiple Sclerosis/ or *Alzheimer Disease/ or parkinsonism/ or exp delirium/ or (delirium$ or delirious$ or Parkinson$ or Alzheimer$).tw. or ((multiple$ or disseminated$) adj5 sclerosis$).tw,kw. (544154)

20 or/16-19 (2199890)

21 (hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-thalam$ or intracranial$ or intra-cranial$).tw,kw. (1452911)

22 ((cortex$ or cortices$ or area$ or region$) adj3 (orbital$ or prefrontal$ or orbitofrontal$)).tw,kw. (77598)

23 ((pars$ adj (triangulari$ or operculari$)) or ((area$ or region$) adj2 (speech$ or language$ or broca$)) or broca's or brocas or subcallosal$).tw,kw. (8645)

24 ((frontal$ or olfactory$) adj3 (sulci or (superior$ adj5 convolution$) or sulcus$)).tw,kw. (877)

25 ((frontal$ or orbital$ or orbitofrontal$ or rectal or rectus or straight or marginal) adj3 (gyri$ or gyrus$)).tw,kw. (16960)

26 or/21-25 (1519587)

27 brain atrophy/ or exp *brain disease/ or (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$).tw,kw. (3600056)

28 26 and 27 (756089)

29 ((brain or cortex$ or cortices$ or cortical$ or subcortical$ or sub-cortical$ or hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-thalam$ or intracranial$ or intra-cranial$) adj2 (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$ or impair$ or decline$ or defect$ or lesion$ or disorder$ or patholog$ or integrity or hyperintens$)).tw,kw. (387036)

30 ((visuospatial$ or (visuo$ adj2 spatial$) or (verbal$ adj2 (comm$ or fluen$)) or (process$ adj2 speed$) or psychomotor$ or psycho-motor$ or memory$ or personality$ or perceptual$ or attention$ or concentration$ or auditory$ or visual$ or sensory$ or sensation$ or spatial$ or IQ or intelligence$ or intellectual$) adj5 (impairment$ or impaired or deficit$ or deficien$ or abnormal$ or dysfunction$ or function$ or skill$ or problem$ of complain$)).tw,kw. (371991)

31 (MMSE or (mini?mental$ adj3 stat$ adj3 exam$)).tw,kw. (24495)

32 mental stress/ or anxiety/ or Depression/ or major depression/ or treatment resistant depression/ or (stress$ or ((hospital or scale$ or score$ or questionnaire$) and (HADS-D or PHQ-9 or PHQ or HADS)) or depress$ or anxiety$).tw,kw. (2068051)

33 posttraumatic stress disorder/ or (shock$ adj2 shell$).tw,kw. or ((posttrauma$ or (post$ adj3 traumatic$) or post-trauma$) adj3 (stress$ or disorder$ or psychiatric$)).tw,kw. or ((acute$ adj2 stress$ adj2 disorder$) and ASD).tw,kw. (69838)

34 ((posttraumatic$ or (post$ adj3 traumatic$) or post-traumatic$ or stress$ or disorder$) and PTSD).tw,kw. (31596)

35 or/32-34 (2079206)

36 exp *blood pressure/ or heart rate/ or heart rate variability/ or respiratory sinus arrhythmia/ or resting heart rate/ or ((heart$ or pulse$) adj3 rate$1).tw. or (cardiac$ adj2 chronotrop$).tw. or (pressure$ adj3 (blood$ or diastolic$ or pulse$ or systolic$)).tw. (799450)

37 (psychiatr$ or disability$).tw,kw. or intellectual impairment/ or exp mental deficiency/ or exp *disability/ or "International Classification of Functioning, Disability and Health"/ (777645)

38 or/36-37 (1568243)

39 20 or 28 or 29 or 30 or 31 or 35 or 38 [Cognitive concept ] (5794698)

40 10 and 39 [ICU concept PLUS volatile anesthetics concept PLUS Cognitive concept ] (1385)

41 survivor/ or surviv$.mp. or (post?intensive$ or postdischarg$ or ((post$ or after$) adj3 (critical$ or ICU or intensive$ or discharg$))).tw,kw. or exp comparative study/ or (vs$1 or vs or versus or compar$ or during-and-after).tw. or ((multicenter$ or multi-center$).tw. and ((between or each or both or either or than or across) adj2 group$1).tw,kw.) or groups.tw. or group$1.ab. /freq=2 (12126728)

42 ((random$ or multicenter$ or multi-center$ or center$ or multicentr$ or multi-centr$ or centr$ or pilot$ or population$ or observational or validation or phase$) adj3 (trial or study)).mp. or clinical trial.tw. or exp clinical study/ or randomization/ (11071495)

43 or/41-42 [ Random/Clinical trial / Survivors / Comparative Study] (18293986)

44 10 and 14 and 43 [WITHOUT Cognitive concept PRESENT could be found in the Full Text - ICU concept PLUS volatile anesthetics concept PLUS Intravenous Anesthesia concept PLUS Random/Clinical trial / Survivors / Comparative Study] (1514)

45 40 or 44 (2262)

46 45 not ((exp animal/ or nonhuman/) not exp human/) (1900)

47 46 not (animal$1 or mice or rat or rats or cat$1 or cattle$1 or dog$1 or goat$1 or horse$1 or rabbit$1 or sheep$1 or swine$1 or pig$1 or piglet* or canine$1 or feline$1 or porcine$ or calf or primate* or rodent$ or hamster$ or lamb$1 or monkey$1 or murine or veterinar*).ti. (1855)

48 limit 47 to (adult <18 to 64 years> or aged <65+ years>) (767)

49 limit 47 to (embryo or infant or child) (326)

50 47 not (49 not (48 and 49)) (1578)

51 50 and adult$.ti. (24)

52 50 and (pediatr$ or paediatr$ or child$ or adolescent$ or infan$ or newborn$ or boy$1 or neonat$).ti. (81)

53 50 not (52 not (51 and 52)) (1500)

54 exp case control study/ or (case$ and control$).tw. or exp case study/ or (case$ and series).tw. [EMBASE case series ] (1136668)

55 case report/ or case report$.tw,kw. (2743289)

56 53 not (55 not (54 and 55)) (1216)

57 56 not case report.ti. [Removing case Reports and retaining Case Series] (1202)

58 limit 57 to english language (1078)

**59 limit 58 to yr="1970 -Current" (1068)**

**60 limit 59 to embase (668)**

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**Database: APA PsycInfo <1967 to November Week 4 2020>**

Search Strategy:

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1 (((inhal* or insufflation$ or gas or gaseous$ or volatile$ or halogen$) adj3 (an?esthe* or sedat*)) or (inhalation$ adj3 induction$) or (ftorotan$ or fluothane$ or narcotan$ or isoflurane$ or aerane$ or aerrane$ or forane$ or forene$ or forthane$ or isoflurano$ or isorane$ or sofloran$ or sevocalm$ or sevoflo$ or sevofrane$ or sevohale$ or sevoflurane$ or sevorane$ or sevotec$ or ultane$ or anestane$

or bromochlorotrifluorethane$ or (bromo$ adj2 chloro$ adj2 trifluorethane$) or fluorothan* or halan$ or halothan* or halthane$ or ineltano$ or phthorothane$ or trothane$ or desflurane$ or enflurane$ or penthrox$)).tw. [volatile anesthetics concept] (1319)

2 intensive care/ (4348)

3 ((intens* or critical*) adj3 (care or ill or illness or unit*)).tw. (13241)

4 (ICU or ICUs or CCU or CCUs or ITU or HDU).mp. (2736)

5 or/2-4 [ICU concept] (14101)

6 (intens$ adj3 care).af. or (intensive* or critical$).jn,jx. or (sedat$.mp. and (*artificial respiration/ or (((ventilat$ or respirat$) adj2 (artificial$ or mechanical$ or pneumon$ or controlled)) or (ventilat$ adj3 patient$)).tw.)) (38989)

7 5 or 6 (42014)

8 1 and 7 (94)

9 intravenous injections/ or (intravenous$ or intra-venous$).mp. or (IV or IVs or TIVA or Auto?TIVA or Manual?TIVA).tw. (76553)

10 Propofol/ or (propofol$ or diprivan$).mp. or Ketamine/ or (ketamin$ or esketamin$ or ketanest$ or ketalar$ or ketaset$).mp. or Midazolam/ or (midazolam$ or dormicum$ or Versed$ or buccolam$ or dalam or dormonid or fortanest or fulsed or hypnovalor hypnovel or hypnoyvel or ipnovel or midacum or midazo or midazol or midolamor miloz).mp. or conscious sedation/ or deep sedation/ or ((conscious$ or moderate$ or deep$) adj2 sedat$).tw. (6565)

11 10 and (continuous$ or prolonged$ or concentration$ or infusion$).mp. (1488)

12 9 or 11 [Intravenous Anesthesia OR Midazolam/Ketamine/Propofol/ continuous administration - concept] (77551)

13 8 and 12 (30)

14 exp Cognition/ or exp Cognitive Therapy/ or Tremor/ or exp neuropsychological assessment/ (69761)

15 exp communication disorders/ or exp memory disorders/ or cognitive impairment/ or exp perceptual disturbances/ (112915)

16 (cognition$ or cognitive$ or neuropsycholog$ or neuro-psycholog$ or neurocognit$ or neuro?affectiv$ or neuro-cognit$ or neurobehav$ or neuro-behav$ or aphasia$ or (involuntary$ adj2 quiver$) or tremor$).mp. (663872)

17 *Multiple Sclerosis/ or *alzheimer's disease/ or parkinsonism/ or delirium/ or (delirium$ or delirious$ or Parkinson$ or Alzheimer$).tw. or ((multiple$ or disseminated$) adj5 sclerosis$).tw. (114306)

18 or/14-17 (780410)

19 (hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-thalam$ or intracranial$ or intra-cranial$).tw. (201667)

20 ((cortex$ or cortices$ or area$ or region$) adj3 (orbital$ or prefrontal$ or orbitofrontal$)).tw. (40292)

21 ((pars$ adj (triangulari$ or operculari$)) or ((area$ or region$) adj2 (speech$ or language$ or broca$)) or broca's or brocas or subcallosal$).tw. (5216)

22 ((frontal$ or olfactory$) adj3 (sulci or (superior$ adj5 convolution$) or sulcus$)).tw. (366)

23 ((frontal$ or orbital$ or orbitofrontal$ or rectal or rectus or straight or marginal) adj3 (gyri$ or gyrus$)).tw. (7911)

24 or/19-23 (238556)

25 (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$).tw. (282352)

26 24 and 25 (66855)

27 ((brain or cortex$ or cortices$ or cortical$ or subcortical$ or sub-cortical$ or hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-thalam$ or intracranial$ or intra-cranial$) adj2 (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$ or impair$ or decline$ or defect$ or lesion$ or disorder$ or patholog$ or integrity or hyperintens$)).tw. (64349)

28 ((visuospatial$ or (visuo$ adj2 spatial$) or (verbal$ adj2 (comm$ or fluen$)) or (process$ adj2 speed$) or psychomotor$ or psycho-motor$ or memory$ or personality$ or perceptual$ or attention$ or concentration$ or auditory$ or visual$ or sensory$ or sensation$ or spatial$ or IQ or intelligence$ or intellectual$) adj5 (impairment$ or impaired or deficit$ or deficien$ or abnormal$ or dysfunction$ or function$ or skill$ or problem$ of complain$)).mp. (179366)

29 (MMSE or (mini?mental$ adj3 stat$ adj3 exam$)).tw. (6045)

30 exp stress/ or exp anxiety/ or exp major depression/ or (stress$ or ((hospital or scale$ or score$ or questionnaire$) and (HADS-D or PHQ-9 or PHQ or HADS)) or depress$ or anxiety$).tw. (637796)

31 exp posttraumatic stress disorder/ or (shock$ adj2 shell$).tw. or ((posttrauma$ or (post$ adj3 traumatic$) or post-trauma$) adj3 (stress$ or disorder$ or psychiatric$)).tw. or ((acute$ adj2 stress$ adj2 disorder$) and ASD).tw. (46884)

32 ((posttraumatic$ or (post$ adj3 traumatic$) or post-traumatic$ or stress$ or disorder$) and PTSD).tw. (32616)

33 or/30-32 (640126)

34 blood pressure/ or heart rate/ or ((heart$ or pulse$) adj3 rate$1).tw. or (cardiac$ adj2 chronotrop$).tw. or (pressure$ adj3 (blood$ or diastolic$ or pulse$ or systolic$)).tw. (39341)

35 (psychiatr$ or disability$).mp. or cognitive impairment/ or disability evaluation/ (428052)

36 or/34-35 (464663)

37 18 or 26 or 27 or 28 or 29 or 33 or 36 [Cognitive concept ] (1633003)

38 8 and 37 [ICU concept PLUS volatile anesthetics concept PLUS Cognitive concept ] (53)

39 38 or 13 (62)

40 39 not (animal$1 or mice or rat or rats or cat$1 or cattle$1 or dog$1 or goat$1 or horse$1 or rabbit$1 or sheep$1 or swine$1 or pig$1 or piglet* or canine$1 or feline$1 or porcine$ or calf or primate* or rodent$ or hamster$ or lamb$1 or monkey$1 or murine or veterinar*).ti. (44)

41 40 and adult$.ti. (1)

42 40 and (pediatr$ or paediatr$ or child$ or adolescent$ or infan$ or newborn$ or boy$1 or neonat$).ti. (5)

43 40 not (42 not (41 and 42)) (39)

44 limit 43 to english language (36)

45 limit 44 to yr="1970 -Current" (36)

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